

The Critical Listener

An Embedding-Based Music Recommender Built on Music Criticism

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Introduction. Algorithmic recommendations on platforms such as Spotify [reduce the diversity of music people actually discover](#). As they optimize for user engagement, recommender systems typically suggest albums and songs that match a listener's existing preferences—another dream-pop record for a dream-pop fan, another chart-adjacent rapper for a hip hop fan—rather than surfacing music that shares similar sensibilities but lives outside the genres a listener already favors. This narrows the scope for musical exploration.

This project applies natural language processing (NLP) to full-text music reviews to build a recommendation engine grounded in *semantic similarity* rather than listening history or genre matching. Instead of recommending music that tightly conforms to a listener's taste profile, our model surfaces albums that critics have described using similar language and aesthetic framing. Our approach is designed for listeners interested in deep listening and willing to venture beyond their typical taste profile. It offers critic-informed suggestions that draw on cultural and aesthetic connections that behavioral data and surface-level audio features cannot capture. In doing so, the recommender enriches everyday music discovery, drives traffic to the underappreciated work of music journalists, and gives music industry institutions and executives an alternative lens on discovery.

Data. The model draws on over 48,000 reviews published between 1999 and 2026 in [Pitchfork](#), [Resident Advisor](#), and [CritiqueBrainz](#). Our review database covers over 44,000 albums across genres such as rock, electronic, hip hop, pop, experimental, and R&B. Band member names mentioned in reviews were masked to avoid matching albums on artist identity rather than descriptive content, prioritizing name-agnostic characterizations of each release.

Recommender. Recommendations are based on the semantic similarity between review text embeddings. For each review, we generate an embedding using an NLP model, then measure pairwise cosine similarity across all reviews and recommend top-n albums whose reviews are the most similar ones. When an album had multiple reviews, we averaged and normalized their embeddings first.

We tested three NLP embedding models against a TF-IDF lexical baseline built on simple word frequency. Because many reviews exceeded a 512-token limit, our comparison included both short- and long-context models. The long-context model Nomic (8,192 token limit) outperformed the TF-IDF baseline most effectively. Since no labeled dataset of “similar albums” exists, and training on same-album pairs would not optimize for the intended task, we deployed Nomic in its zero-shot rather than fine-tuned form.

Because no ground-truth “correct similarity” labels exist for album recommendation outside of user listening data, we could not rely on traditional validation methods. This was addressed in two ways:

Explainer. First, we developed an LLM-based “explainer” that reads reviews of a seed album and its recommended counterparts, identifies genuine shared qualities, and presents both reviews side by side so the shared thematic and aesthetic ground is visible to the user. Every connection is grounded in verbatim quotes from real reviews, making the system's reasoning auditable and preparing the listener for what they are about to hear. After spot-checking dozens of queries by hand, we concluded that our model's recommendations reliably surface genuine thematic connections, providing strong qualitative evidence that the approach works as intended. An example using Drake's popular album *Take Care* (2011) is provided as an appendix to this document.

Benchmark. Second, we benchmarked our recommendations against an industry-style collaborative filtering recommender. Using the [Last.fm](#) API, we generated 45,000 recommendations for 11,000 seed albums, incorporating metadata such as listener counts and user-generated tags, and then computed 30+ metrics to evaluate recommendation quality. Measures were calculated at album-level, as well as at artist- and tag-level, and grouped into themes, such as disagreement/overlap, popularity bias, diversity, repetition/hubness, novelty, and serendipity. Collectively, they provided a way to compare our approach against collaborative filtering on dimensions that matter for discovery, not just prediction accuracy. Below we present highlights of our analysis, with a full report available in [this notebook](#).

To what extent do our recommendations meaningfully differ from industry-style outputs? Disagreement was measured using Jaccard similarity, which captures the overlap between the albums, artists, and tags each recommender produces for a seed album. As our analysis shows, mean scores were low across all categories, indicating that the model produced meaningfully different recommendations. Some overlap still existed, however—12% of queries had at least one artist overlap, while 95% had one or more matching tags—reflecting that our recommendations retained some alignment with the baseline.

Do our recommendations avoid common pitfalls of recommenders, such as over-suggesting a narrow set of popular items or defaulting to “safe” picks with little discovery value? A few remarks:

- **Lower popularity bias:** Our model’s recommendations averaged 225,000 listeners per album versus 293,000 for Last.fm (a 23% reduction). They were also 60 times more popular than the query album versus 111 times for the baseline, showing that our model outperforms Last.fm in moving further beyond obvious picks.
- **Greater diversity in artists and tags:** On average, our model’s top five recommendations featured five distinct albums from different artists in 62% of queries versus 41% for Last.fm, with fully distinct tags appearing in 39% of lists versus 23%, giving users a wider range of options.
- **More repetition than the baseline:** heavily recommended “hub” albums filled 87% of slots versus 44% for Last.fm, meaning the model recommends fewer unique albums overall. This gap is expected given the difference in catalog size (roughly 44,000 albums in our corpus versus Last.fm’s 2.2 million).

A qualitative analysis of Drake’s *Take Care* example captures these differences well: whereas an industry recommender centers on chart-adjacent, high-visibility peers, our model’s picks are less predictable, less concentrated among a handful of popular artists, and more varied in aesthetic and cultural reference points, while remaining demonstrably grounded in shared critical language.

Conclusion. This project demonstrates that NLP-based music recommenders built on semantic similarity between album reviews can produce more unexpected and diverse suggestions than industry models. Compared with industry standards, our model offers meaningfully different, lower-popularity-bias, higher-diversity suggestions. Together with our auditable, quote-grounded explainer of why each recommendation was made, our recommender gives curious listeners a path to discovery that behavioral or audio-feature-based systems do not provide. Its effectiveness is currently limited by reliance on only three review sources, an occasional tendency to base recommendations on marginally shared qualities, and a lack of more robust quantitative validation metrics using expanded metadata. Future iterations could incorporate larger review datasets and integrate more complex guidelines that more effectively prevent superficial or irrelevant similarities.

Appendix

Analysis of our model's recommendations for Drake - *Take Home* (2011)

Top 3 Recommendations

Our Model's Recommendations	Last.fm Model's Recommendations
Stwo - <i>D.T.S.N.T</i> French electronic/chillwave producer known for hazy, atmospheric instrumentals	J. Cole - <i>Cole World: The Sideline Story</i> Mainstream conscious hip hop debut from a chart-topping rapper
Majid Jordan - <i>Majid Jordan</i> Canadian R&B/synth-pop duo blending dreamy production with dance-pop textures	Drake & 21 Savage - <i>Her Loss</i> Collaborative mainstream rap album co-starring Drake himself
Partynextdoor - <i>Partynextdoor</i> Moody alt-R&B project built on minimalist, late-night production	Kanye West - <i>Yeezus</i> Top-charting hip hop album from an A-list rapper

The "Explainer"

Similarities between Drake - *Take Care* & Stwo - *D.T.S.N.T.*

Thematic Similarity	Citations from Reviews
Production aesthetic of muted, atmospheric backdrops	"its muted backdrops - like listening to a record on your neighbour's stereo, through the dividing wall - were perfect for emoting over." (<i>Take Care</i> , <i>CritiqueBrainz Review</i>)
	"The drums are programmed and flat, the mix is smoky and these beautiful voices float atop it all, creating a drugged-out haze" (<i>D.T.S.N.T.</i> , <i>Pitchfork Review</i>)
Collaboration with skilled producers to elevate artistic vision	"...by surrounding himself with the right producers and guests (and presumably the right management), he's elevated himself to remarkable heights" (<i>Take Care</i> , <i>CritiqueBrainz Review</i>)
	"STWO...is worth seeking out if, like Drake, you're a vocalist looking to flex your abilities. On his new EP, D.T.S.N.T., he makes room for a globally sourced array of collaborators..." (<i>D.T.S.N.T.</i> , <i>Pitchfork Review</i>)
Noah "40" Shebib's production influence	"Calling predominantly on the production skills of Noah '40' Shebib..." (<i>Take Care</i> , <i>CritiqueBrainz Review</i>)
	"STWO's collaborators thrive over the kinds of beats that producers like Noah '40' Shebib, Clams Casino, and Shlohmo mastered long ago" (<i>D.T.S.N.T.</i> , <i>Pitchfork Review</i>)